AI-powered digital safety and phishing detection

"Making Digital Safety Understandable: AI-Powered Detection and Guidance for Phishing, Scams and Suspicious Digital Content. give me the exact details so i can do it very fast" *Generated by LeapSpace*

Making Digital Safety Understandable at a Glance

Key Findings Table

Key finding	Relevant setting / users	Why it matters for fast implementation	Citations
High benchmark accuracy does not guarantee real-world safety	Email, URL, SMS, social and cross-dataset deployment	You should optimize for cross-dataset performance, calibration, and realistic prevalence, not only random-split accuracy	[1] [2] [3] [4] [5] [6] [7]
Tree ensembles and transformers are the strongest practical model families	Email phishing, URL phishing, multilingual detection	Random Forest/GBM are strong deployable baselines; BERT/RoBERTa/XLM-R often win static benchmarks	[8] [9] [10] [11] [12] [13] [14]
False-positive control is critical for user trust	Inbox filtering, browser warnings, real-time alerting	A strong detector that warns too often becomes unusable; thresholds and calibration are core product decisions	[5] [15] [16] [17] [18] [19] [20]
Understandable guidance must go beyond "suspicious" labels	Consumer and employee-facing warnings	Users respond better when warnings explain why and what to do next	[21] [22] [23] [24] [25]
Warning design has mixed effects	Browser/mobile/payment/security warnings	Icons, wording, confidence cues, and banner placement can help or backfire, so warning UX must be tested	[26] [27] [28] [29]
Multimodal detection is becoming the new standard	Webpages, spoofed login pages, brand impersonation	Text alone is not enough; best systems combine URL, screenshot, DOM, logo, and layout cues	[30] [31] [32] [33]
Multilingual coverage is now practical with transformer backbones	English and non-English phishing/misinformation	XLM-R-style models enable regional-language scam detection without relying only on translation	[13] [14] [34] [35]

Key finding	Relevant setting / users	Why it matters for fast implementation	Citations
Human behavior remains the limiting factor	SMEs, healthcare, enterprise users, consumers	Detection must be paired with training, nudges, and just-in-time guidance	[36] [37] [38] [39] [40] [41] [42] [43]

Direct Answer

If you want to build this **very fast**, the strongest evidence supports a **layered AI safety system** with five core parts: **(1)** a detector using **text + URL + page/screenshot features**, **(2)** a **calibrated risk threshold** tuned for low false positives, **(3)** a **plain-language reason** for the warning, **(4) one-click next actions** such as "do not click," "verify sender," "open the official site separately," and "report/block," and **(5)** testing on **cross-dataset, multilingual, and adversarial cases**, not just on clean benchmark splits [14] [18] [23] [25] [31] [32] [44].

For the detection layer, use **Random Forest or Gradient Boosting as practical baselines** and **BERT/RoBERTa/XLM-R** where you need higher static performance or multilingual coverage, because those model families consistently lead across email, URL, and multilingual deception tasks, though no single architecture dominates every context [9] [10] [11] [12] [13] [14].

For the user-facing layer, do **not** show only a binary "danger" badge. The evidence favors **short textual explanations**, **high visual salience**, and **step-by-step guidance**, but also shows that some warnings can be ignored or even increase risky behavior, so the exact alert design must be validated with users [22] [24] [26] [28] [29].

Main caution: the literature is much stronger on **detection accuracy** than on phishing-specific proof that **AI explanations directly improve user understanding and safer behavior in production**, so your product should treat explanation UX as something to test, not assume [45] [46] [47].

Confidence level: High - this answer aligns with repeated cross-source findings on model choice, deployment realism, warning design, and human factors [5] [10] [24] [25] [31].

Study Scope

This report synthesizes evidence from the supplied literature on **2019-2025 AI phishing/scam detection**, plus foundational pre-2020 benchmark and adversarial-classification work where it shaped current practice [48] [49] [50] [51]. It covers **email, SMS, Telegram/messaging, URLs, phishing webpages, browser settings, multilingual text, multimodal brand impersonation, warning design, explainable AI, robustness, and human behavior** [14] [18] [26] [32] [40] [52] [53].

Assumptions & Limitations

- Many supplied studies report **accuracy** more often than complete **precision/recall/F1/FPR/FNR** tables, so exact operational comparisons are incomplete [54] [55] [56].

- Phishing-specific **XAI user studies** are sparse; many explanation claims come from adjacent security or AI decision-support settings rather than direct phishing-warning trials [45] [46] [57] [58].
- Phishing-specific **adversarial robustness reporting** is limited; the risk is widely acknowledged, but exact attack success rates are often missing [44] [59] [60] [61].
- Operational details such as **latency, memory, and privacy design** are discussed unevenly across papers, especially in SMS and messaging settings [62] [63] [64].

Suggested Further Research

- Run phishing-specific user studies testing whether explanations improve **comprehension, speed, trust, and safe behavior** [45] [46].
- Standardize evaluations combining **random split, temporal split, cross-dataset transfer, and prevalence shift** in one benchmark [5] [7].
- Report phishing-specific robustness against **LLM rewriting, URL obfuscation, OCR/logo spoofing, and poisoning** [33] [44] [60].
- Expand **low-resource multilingual and multimodal scam datasets** for SMS, chat, and voice-text hybrids [14] [35] [65].
- Quantify the behavioral and economic cost of **false positives** and alert fatigue in deployed systems [18] [19] [20].

Full Report

Foundational Context

Digital safety systems for phishing, scams, and suspicious content have two separate jobs: **detect the threat** and **make the risk understandable enough that a user acts safely** [66] [67] [68]. The first job is a classification problem across channels such as email, SMS, URLs, webpages, and social content [1] [30] [53]. The second job is a human-decision problem shaped by trust, attention, wording, and perceived credibility [22] [25] [29] [69].

The supplied evidence shows a persistent gap between these jobs. Many systems report very high offline performance, but deployment success depends on **generalization, calibration, low false positives, latency, and user behavior** rather than peak benchmark accuracy alone [4] [5] [9] [18]. Taken together, these findings suggest that a usable digital-safety product should be designed as a **decision-support system**, not only a classifier [23] [25].

Confidence level: High - multiple meta-insights and domain-specific findings support this framing [1] [4] [23] [25].

1. Threat Landscape: Why Detection Must Be Understandable

Phishing and scams now span **email, SMS, messaging apps, spoofed webpages, social platforms, and multimodal impersonation content** [30] [41] [70] [71]. The research also describes a shift toward **AI-assisted attacks** that are more personalized, fluent, scalable, and persuasive than traditional templates [41] [42] [43] [63].

That changes the product requirement. Detection alone is insufficient because users still decide whether to click, reply, submit credentials, or override the warning [1] [4] [25] [72]. Human vulnerability is linked to trust, stress, knowledge gaps, impulsivity, familiarity with the sender, and organizational context [36] [37] [38] [73] [74]. In enterprise and consumer settings, even effective technical controls can fail if the warning is confusing, mistimed, or not trusted [25] [28] [75].

2. AI Detection Models for Phishing and Scam Content

2.1 Email phishing detection

Across the supplied email studies, classic models already perform strongly offline: Naive Bayes reached **96% email** and **97% URL** accuracy in one study, Random Forest reached **98.64%** on phishing URLs, and transformer models such as BERT/RoBERTa reached about **99%** on merged email datasets [1] [8] [9]. More recent direct comparisons report that **BERT/RoBERTa or DistilBERT generally outperform Naive Bayes, Random Forest, and SVM**, while **hybrid pipelines** that add structural features can outperform pure text transformers and handle temporal drift better [9] [10] [76] [77].

Among the classical baselines, **Random Forest repeatedly emerges as the strongest general-purpose option**, with Naive Bayes and SVM remaining competitive but usually weaker [10] [78] [79]. One optimized Naive Bayes pipeline with hybrid feature engineering reportedly improves false-positive behavior relative to conventional Naive Bayes, which matters when the model is exposed directly to users [56].

The main implementation lesson is that **architecture alone is not decisive**. Dataset quality, feature engineering, and drift handling repeatedly matter as much as model family [2] [46] [77] [80].

2.2 SMS and messaging-platform detection

SMS spam research commonly relies on the **UCI SMS Spam Collection**, including a cited configuration with **5,574 English messages**, while another study notes **13.4% spam prevalence**, highlighting class imbalance as a design issue [81] [82]. A weighted soft-voting ensemble improved recall over Naive Bayes on SMS data [81]. Telegram spam detection on **20,348 messages** found **ALBERT** and **GRU** strongest for the minority spam class, with ALBERT reaching **F1 0.95** and GRU **F1 0.90** [52].

For deployment realism, the literature emphasizes dataset authenticity, metadata coverage, and balance, but true cross-dataset messaging evaluations remain scarce [3]. Near-real-time SMS classification has been demonstrated using a **Flask API** and **Twilio-based intake**, though precise latency is often not reported [83] [84].

2.3 URL-, host-, and content-based phishing classifiers

URL and webpage phishing detection is one of the most mature areas. Systems compare **lexical URL features, host/WHOIS signals, HTML/JavaScript content, text, images, and hybrid signals** with tree models, boosting methods, and deep learning [12] [53] [85] [86]. Ensemble and tree-based models are repeatedly among the strongest, with **GBM** often described as best overall and **Random Forest** a strong competitor [11] [12] [87] [88].

One especially practical finding is that combining **lexical URL features with site content and image analysis** can reduce false positives in real-time trials, which is more valuable for product use than headline benchmark accuracy alone [89].

2.4 Foundational benchmark works

Pre-2020 studies established many of today's data and evaluation norms. Foundational works used feature-rich email datasets, lexical and host-based URL features, content TF-IDF, anomaly-based discrepancy analysis, and time-based evaluation rather than only randomized splits [48] [90] [91] [92] [93] [94]. CANTINA+ and blacklist measurement studies helped establish the importance of **fresh-phish performance**, **false positives**, and **time-sensitive evaluation** [49] [92].

Confidence level: High - there is consistent evidence across email, SMS, URL, and benchmark literature on dominant model families and the importance of features and evaluation design [10] [11] [52] [92].

3. Explainable Warnings and User Guidance

3.1 What explanations can contribute

Across security applications, XAI is valued because it can show **confidence scores, suspicious words, obfuscated URLs, and feature-level reasons** that help humans decide what to do [21] [22] [23]. Reviews argue that the real value of explainability is not only transparency but **actionability** and **faster, more accountable decision-making** [66] [67] [68] [95].

However, a key evidence gap is that **no retrieved paper directly tests phishing/scam XAI with user-study outcomes such as understanding, trust, or decision speed** [45] [46]. SHAP and LIME are frequently highlighted in the broader cybersecurity literature, and counterfactuals are described as intuitive for human reasoning, but these claims are mostly indirect for phishing [67] [96] [97] [98].

3.2 Warning design and trust

Warning design clearly changes behavior, but not always in a linear way. **Text

References

1. Dhanalakshmi, B., Rajeshwari, R.R., Sanju, A.N., (...), Venkatesan, P. (2026) Email Spam Filtering Using Artificial Intelligence Techniques. *Diyala Journal of Engineering Sciences*. https://doi.org/10.24237/djes.2026.19102
2. García, N., Sotomayor, J., Ticona, W. (2026) Efficiency of Deep Learning Models in Phishing Detection on Social Networks: A Systematic Review. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-032-22091-2_16
3. Johari, M.F., Chiew, K.L., Hosen, A.R., (...), Grzonka, D. (2025) Key insights into recommended SMS spam detection datasets. *Scientific Reports*. https://doi.org/10.1038/s41598-025-92223-1
4. Yıldırım, M. (2026) Adversarial Robustness in URL-Based Phishing Detection: Problem-Space Evaluation and Robust Feature Engineering. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app16178737
5. Bhuiyan, I., Bhuiyan, T. (2026) The Fragility of Phishing Detection Models: Evidence from Cross-Corpus Transfer, Prevalence Shift, Artifact Learning, and Evasion Risk. *Big Data and Cognitive Computing*. https://doi.org/10.3390/bdcc10070211
6. Misiek, M., Hyla, T. (2026) XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation. *IEEE Access*. https://doi.org/10.1109/ACCESS.2026.3672690
7. Adinarayana, R., Krishna, G.V. (2026) Cross-Dataset Generalization Framework for Cybercrime Detection Using CIC-IDS2017, CIC-Phishing2019, and Malicious Uniform Resource Locator (URL) Data. *Journal of Advances in Information Technology*. https://doi.org/10.12720/jait.17.7.1310-1320
8. Aniyansyah, N., Rina, R., Puspitasari, S., Erfina, A. (2025) Evaluation of AI Models for Phishing Detection Using Open Datasets †. *Engineering Proceedings*. https://doi.org/10.3390/engproc2025107037
9. Alhuzali, A., Alloqmani, A., AlJabri, M., Alharbi, F. (2025) In-Depth Analysis of Phishing Email Detection: Evaluating the Performance of Machine Learning and Deep Learning Models Across Multiple Datasets. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app15063396
10. Tarapiah, S., Abbas, L., Mardawi, O., (...), Mansoor, W. (2025) Evaluating the Effectiveness of Large Language Models (LLMs) Versus Machine Learning (ML) in Identifying and Detecting Phishing Email Attempts. *Algorithms*. https://doi.org/10.3390/a18100599
11. https://www.sciencedirect.com/science/article/pii/S2624909X2500102X
12. Zhai, J., Wang, X., Zhai, Z., (...), Yang, H. (2025) Industrial IoT intrusion attack detection based on composite attention-driven multi-layer pyramid features. *Computer Networks*. https://doi.org/10.1016/j.comnet.2025.111207
13. Staples, D., Hakak, S., Cook, P. (2023) A Comparison of Machine Learning Algorithms for Multilingual Phishing Detection. *2023 20th Annual International Conference on Privacy, Security and Trust, PST 2023*. https://doi.org/10.1109/PST58708.2023.10320177

14. Sanjay, S., Krishna, N., Paramesh, G., Kiran Kumar, K.V. (2026) An LLM-Based Multilingual Phishing Detection Framework for Regional Languages. *International Conference on AI-Driven Innovations and Applications, IC-AIDA 2026 - Conference Proceedings*. https://doi.org/10.1109/IC-AIDA68291.2026.11564455

15. Sultan, D., Kamalov, D., Sadu, S., (...), Kokenov, R. (2026) AI-Powered Phishing Attacks and How to Prevent Them: A Comprehensive Survey and Framework. *2026 International Conference on Cybersecurity, Digital Forensics, and AI Applications, ICCSDFAI 2026*. https://doi.org/10.1109/ICCSDFAI70505.2026.11648063

16. Pavithra, J., Selvakumara Samy, S. (2022) A Comparative Study on Detection of Malware and Benign on the Internet Using Machine Learning Classifiers. *Mathematical Problems in Engineering*. https://doi.org/10.1155/2022/4893390

17. Basha, M.E.S., Rao, M.N. (2026) Multi-Source Generalization-Aware Phishing URL Detection Using Calibrated Stacked Ensemble and False-Positive Control. *Journal of Trends in Computer Science and Smart Technology*. https://doi.org/10.36548/jtcsst.2026.3.020

18. Shi, F. (2026) Usable Phishing Defense in the Browser via Multi-Stage Detection and Trust Controls. *2026 IEEE 4th International Conference on Control, Electronics and Computer Technology, ICCECT 2026*. https://doi.org/10.1109/ICCECT68671.2026.11565112

19. Alhasan, S.A.M., Al-Allawee, A., Ali, A.K. (2026) EmailGuard: A Multi Agent Ensemble based Multi-modal Email Phishing Detection Framework with SDN Enforcement. *International Journal of Intelligent Engineering and Systems*. https://doi.org/10.22266/ijies2026.0831.63

20. Vicente, Lucía, Matute, Helena (2026) Warning people about the risk of AI error mitigates human acquisition of AI bias. *Cognitive Research: Principles and Implications*. https://doi.org/10.1186/s41235-026-00726-w

21. Gangrade, M., Sivasamy, S. (2026) Demystifying Phishing Detection: Using Explainable AI to Identify Suspicious Email Features. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-032-25631-7_25

22. Bezzaoui, I., Stein, C., Weinhardt, C., Fegert, J. (2025) Explainable AI for online disinformation detection: Insights from a design science research project. *Electronic Markets*. https://doi.org/10.1007/s12525-025-00799-3

23. Patidar, M., Panthi, S., Zalawadia, K. (2025) A Systematic Review of Explainable Machine Learning Approaches in Phishing URL Detection. *IEEE International Conference on Innovate for Humanitarian: Tech Solutions for Global Challenge, ICIH 2025*. https://doi.org/10.1109/ICIH67754.2025.11608401

24. https://www.sciencedirect.com/science/article/pii/S1573761625000390

25. Wang, Y., Qin, M., Tao, D., Liu, Z. (2026) The Role of Explanation Format and Certainty Expression in User Trust Towards AI-Assisted Healthcare Recommendations. *Lecture Notes in Computer Science*. https://doi.org/10.1007/978-3-032-30047-8_8

26. Grummon, A.H., Ruggles, P.R., Greenfield, T.K., Hall, M.G. (2023) Designing Effective Alcohol Warnings: Consumer Reactions to Icons and Health Topics. *American Journal of Preventive Medicine*. https://doi.org/10.1016/j.amepre.2022.09.006

27. Wang, J., Shi, J., Wen, X., (...), Qian, X. (2022) The effect of signal icon and persuasion strategy on warning design in online fraud. *Computers and Security*. https://doi.org/10.1016/j.cose.2022.102839

28. Junger, M., Montoya, L., Overink, F.-J. (2017) Priming and warnings are not effective to prevent social engineering attacks. *Computers in Human Behavior*. https://doi.org/10.1016/j.chb.2016.09.012

29. Hancock, P.A., Kaplan, A.D., Macarthur, K.R., Szalma, J.L. (2020) How effective are warnings? A meta-analysis. *Safety Science*. https://doi.org/10.1016/j.ssci.2020.104876

30. Vulfin, A., Sulavko, A., Vasiliev, V., (...), Samotuga, A. (2026) A Multimodal Phishing Website Detection System Using Explainable Artificial Intelligence Technologies. *Machine Learning and Knowledge Extraction*. https://doi.org/10.3390/make8010011

31. Sophia, B., Samritha, S., Dhanalakshmi, M., Indrapriyadharshini, T. (2025) Multimodal Phishing Detection Using Graph Neural Networks, BERT, and CLIP. *3rd International Conference on Emerging Research in Computational Science, ICERCS 2025*. https://doi.org/10.1109/ICERCS65898.2025.11580482

32. Ganesh Kumar, K., Devi S, V., Vaishnavi, T., (...), Keerthana, N.V. (2025) Phishing Website Detection Using a Hybrid CNN-CapsNet-XGBoost-Random Forest Model for Cross-Platform Security. *2025 2nd Asia Pacific Conference on Innovation in Technology, APCIT 2025*. https://doi.org/10.1109/APCIT65661.2025.11411149

33. Lee, J., Lim, P., Hooi, B., Divakaran, D.M. (2024) Multimodal Large Language Models for Phishing Webpage Detection and Identification. *eCrime Researchers Summit, eCrime*. https://doi.org/10.1109/eCrime66200.2024.00007

34. Chalehchaleh, R., Farahbakhsh, R., Crespi, N. (2024) Multilingual Fake News Detection: A Study on Various Models and Training Scenarios. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-031-66428-1_5

35. Ye, Y., Chen, W., Liu, H., Lun, Z. (2026) Multilingual misinformation detection in social media: A structured survey of challenges and deep learning solutions. *Applied Soft Computing*. https://doi.org/10.1016/j.asoc.2026.116253

36. Duman, S.A., Duman, A., Hayran, R., Soğukpinar, I. (2025) A Multi-Layer Model of Psychological Factors and Parametric Approaches for Human-Centric Phishing Prevention. *ISDFS 2025 - 13th International Symposium on Digital Forensics and Security*. https://doi.org/10.1109/ISDFS65363.2025.11012100

37. Arévalo, D., Valarezo, D., Fuertes, W., (...), MacAs, M. (2023) Human and Cognitive Factors involved in Phishing Detection. A Literature Review. *Proceedings - 2023 Congress in Computer Science, Computer Engineering, and Applied Computing, CSCE 2023*. https://doi.org/10.1109/CSCE60160.2023.00105

38. Abaker, A.A., Alzahrani, I.R., Alharbi, N.A.M. (2027) Exploring the perceived awareness–behaviour gap in AI-enabled phishing: a socio-technical perspective informed by complex adaptive systems. *Computers and Security*. https://doi.org/10.1016/j.cose.2026.105192

39. Roghanizad, M., Choi, E., Mashatan, A., Turetken, O. (2021) Mindfulness and phishing email detection. *27th Annual Americas Conference on Information Systems, AMCIS 2021*. https://www.scopus.com/pages/publications/85118687847

40. Nicholson, J., Coventry, L., Briggs, P. (2017) Can we fight social engineering attacks by social means? Assessing social salience as a means to improve phish detection. *Proceedings of the 13th Symposium on Usable Privacy and Security, SOUPS 2017*. https://www.scopus.com/pages/publications/85075919770

41. Singh, J. (2025) Navigating AI-Enhanced Social Engineering: Strategies for Countering Modern Threats. *International Conference on Software, Systems and Information Technology, SSITCON 2025*. https://doi.org/10.1109/SSITCON66133.2025.11342253

42. Alkamli, F., Qureshi, R.J. (2026) Spear Phishing in Social Engineering: Leveraging ChatGPT and Numberbook. *International Journal of Wireless and Microwave Technologies*. https://doi.org/10.5815/ijwmt.2026.02.05

43. Baidya, S., Mahmud, S., Kabeer, F., (...), Salarian, H. (2026) Social Engineering and Cybersecurity: A Review on Recent Advances in Attack Landscape and Mitigation Strategies. *Lecture Notes in Electrical Engineering*. https://doi.org/10.1007/978-3-032-20235-2_13

44. Rączka, G., Plichta, A. (2026) Spam and Phishing Detection with Adversarial Attack Robustness Analysis Using Machine Learning Models. *Proceedings - European Council for Modelling and Simulation, ECMS*. https://www.scopus.com/pages/publications/105042236606

45. Kritika, E. (2025) A comprehensive literature review on phishing URL detection using deep learning techniques. *Journal of Cyber Security Technology*. https://doi.org/10.1080/23742917.2024.2378552

46. Alghenaim, M., Alkawsi, G., Barnhart, C.R. (2025) The State of the Art in AI-Based Phishing Detection: A Systematic Literature Review. *Studies in Computational Intelligence*. https://doi.org/10.1007/978-3-031-75091-5_23

47. Desolda, G., Aneke, J., Ardito, C., (...), Costabile, M.F. (2023) Explanations in warning dialogs to help users defend against phishing attacks. *International Journal of Human Computer Studies*. https://doi.org/10.1016/j.ijhcs.2023.103056

48. Fette, I., Sadeh, N., Tomasic, A. (2007) Learning to detect phishing emails. *16th International World Wide Web Conference, WWW2007*. https://doi.org/10.1145/1242572.1242660

49. Sheng, S., Wardman, B., Warner, G., (...), Zhang, C. (2009) An empirical analysis of phishing blacklists. *6th Conference on Email and Anti-Spam, CEAS 2009*. https://www.scopus.com/pages/publications/84904793179

50. Dalvi, N., Domingos, P., Mausam, n., (...), Verma, D. (2004) Adversarial classification. *KDD-2004 - Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. https://doi.org/10.1145/1014052.1014066

51. Barreno, M., Nelson, B., Sears, R., (...), Tygar, J.D. (2006) Can machine learning be secure?. *Proceedings of the 2006 ACM Symposium on Information, Computer and Communications Security, ASIACCS '06*. https://doi.org/10.1145/1128817.1128824

52. Xu, S., Ding, Z., Wei, Z., (...), Wang, H. (2025) A Comparative Analysis of Deep Learning and Machine Learning Approaches for Spam Identification on Telegram. *2025 6th International Conference on Computer Communication and Network Security, CCNS 2025*. https://doi.org/10.1109/CCNS66849.2025.11338070

53. Ahmed, Abdulghani Ali, Al-Bayatti, Ali, Saif, Mubarak, (...), Rassem, Taha H. (2023) A Honeybee-Inspired Framework for a Smart City Free of Internet Scams. *Sensors (Basel, Switzerland)*. https://doi.org/10.3390/s23094284

54. Johnston, M., Rahouti, M., Ghani, N., Hayajneh, T. (2025) Optimizing Phishing Detection: Comparative Analysis of Lightweight Machine Learning and Transformer Models. *Proceedings - 2025 IEEE World Forum on Public Safety Technology, WF-PST 2025*. https://doi.org/10.1109/WF-PST65083.2025.00036

55. Hasan, M., Nahar, N., Imam, M.H., Khandaker, M.U. (2025) PhisNet: Intelligent Detection of Phishing. *Eurasia Proceedings of Science, Technology, Engineering and Mathematics*. https://doi.org/10.55549/epstem.1357

56. Vishwa, R., Kamalin, D., Vanaja, S., (...), Sabarinathan, G. (2025) Enhanced Naïve Bayes Model for Intelligent and Secure Email Spam Detection Using Hybrid Feature Engineering and Blockchain-Based Verification. *2025 International Conference on Metaverse and Current Trends in Computing, ICMCTC 2025*. https://doi.org/10.1109/ICMCTC62214.2025.11196588

57. Lundberg, S.M., Lee, S.-I. (2017) A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*. https://www.scopus.com/pages/publications/85044542379

58. Ribeiro, M.T., Singh, S., Guestrin, C. (2016) "Why should i trust you?" Explaining the predictions of any classifier. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. https://doi.org/10.1145/2939672.2939778

59. Hassan, Md.R., Nabin, J.A., Toyeer-E-Ferdoush (2026) Cross-Domain Adversarial Robustness and Explainability Stability in Phishing Detection Systems. *2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure, PECCII 2026*. https://doi.org/10.1109/PECCII70991.2026.11661863

60. Jamil, E., Garcia-Blas, J., Alawadi, S., Carretero, J. (2025) A Deep Experimental Study of Ensemble-Based Phishing Detection in Centralised and Federated Settings. *Proceedings - 2025 IEEE Conference on Pervasive and Intelligent Computing, PICom 2025*. https://doi.org/10.1109/PICom68402.2025.00025

61. Kalyan, K.S. (2024) A survey of GPT-3 family large language models including ChatGPT and GPT-4. *Natural Language Processing Journal*. https://doi.org/10.1016/j.nlp.2023.100048

62. Silva, B., Leite, P., Ribeiro, O.R. (2025) Evaluating Spam Detection Techniques: A Comparison of TF-IDF and Sentence Embeddings with Machine Learning Models. *ISDFS 2025 - 13th International Symposium on Digital Forensics and Security*. https://doi.org/10.1109/ISDFS65363.2025.11012070

63. Aladwan, S.A., Abujazar, I., Alkhannous, M., Houhamdi, Z. (2025) Data Mining for Social Engineering and Phishing Detection: A Systematic Review and Taxonomy (2020-2025). *International Conference on Social Networks Analysis, Management and Security, SNAMS*. https://doi.org/10.1109/SNAMS67467.2025.11391232

64. Islam, M.M., Salsabil, U., Mekhriddin, M., Hossain, S. (2026) Lightweight Intrusion Detection Systems for IoT–Edge Environments: A PRISMA-ScR Systematic Review of Deployability Evidence and a Unified Assessment Framework. *Future Internet*. https://doi.org/10.3390/fi18060300

65. Hamed, H., Hefny, H.A. (2026) Deep-fake detection of digital misinformation over social media: a comprehensive review. *Array*. https://doi.org/10.1016/j.array.2026.101086

66. Wadhwa, K., Babbar, H., Rani, S. (2025) Explainable artificial intelligence in threat detection. *Explainable Artificial Intelligence (XAI) for Next Generation Cybersecurity: Concepts, Challenges and Applications*. https://doi.org/10.1049/PBSE027E_ch3

67. Naeem, M.R., Farhan, M. (2025) Bridging the gap: explainable AI in threat detection and cybersecurity. *Explainable Artificial Intelligence (XAI) for Next Generation Cybersecurity: Concepts, Challenges and Applications*. https://doi.org/10.1049/PBSE027E_ch2

68. Ahmed, R., Hassan, H., Saleem, Z., (...), laghari, A.A. (2026) Explainable AI for Context-Aware Threat Detection. *Unleashing Machine Learning for Advanced Threat Detection and Prevention in Industry 4.0*. https://www.scopus.com/pages/publications/105041175895

69. Ha, T., Kim, S. (2024) Improving Trust in AI with Mitigating Confirmation Bias: Effects of Explanation Type and Debiasing Strategy for Decision-Making with Explainable AI. *International Journal of Human-Computer Interaction*. https://doi.org/10.1080/10447318.2023.2285640

70. Ambrozie, A.R., Cristina, B.A., Scripcariu, S. (2024) AI and Prompt Engineering: The New Weapons of Social Engineering Attacks. *Proceedings of the 16th International Conference on Electronics, Computers and Artificial Intelligence, ECAI 2024*. https://doi.org/10.1109/ECAI61503.2024.10607415

71. Ayodele, T.O., Andritsch, J., Olabanji, D. (2025) Detection and Prevention of Generative AI Email Phishing Attacks Using Digital Twins. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-3-032-00071-2_40

72. Nadeem, A., Vos, D., Cao, C., (...), Verwer, S. (2023) SoK: Explainable Machine Learning for Computer Security Applications. *Proceedings - 8th IEEE European Symposium on Security and Privacy, Euro S and P 2023*. https://doi.org/10.1109/EuroSP57164.2023.00022

73. Sarno, D.M., Harris, M.W., Black, J. (2023) Which phish is captured in the net? Understanding phishing susceptibility and individual differences. *Applied Cognitive Psychology*. https://doi.org/10.1002/acp.4075

74. Tolsdorf, J., Langer, D., Lo Lacono, L. (2025) Phishing Susceptibility and the (In-)Effectiveness of Common Anti-Phishing Interventions in a Large University Hospital. *CCS 2025 - Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*. https://doi.org/10.1145/3719027.3765164

75. Hua, Z., Lu, Y., Li, Q. (2026) When AI warns consumers: source credibility and purchase decisions in ultra-low-priced tourism. *Asia Pacific Journal of Marketing and Logistics*. https://doi.org/10.1108/APJML-05-2026-1359

76. Asliyuksek, H., Tonkal, O., Kocaoğlu, R. (2025) A Comparative Evaluation of a Multimodal Approach for Spam Email Classification Using DistilBERT and Structural Features. *Electronics (Switzerland)*. https://doi.org/10.3390/electronics14193855

77. Jadhav, P., Kadam, Y., Khade, O., (...), Solunke, P. (2025) Empirical Evaluation of Machine Learning Algorithms and T5 Language Model for Advanced Phishing Email Detection. *Lecture Notes in Networks and Systems*. https://doi.org/10.1007/978-981-96-3797-3_8

78. Haikal, H.W.T., Utomo, R.G. (2025) A Comparative Study of Phishing Link Detection Algorithms Using Hybrid and Non-Hybrid Approaches. *International Conference on Information and Communications Technology, ICOIACT*. https://doi.org/10.1109/ICOIACT67584.2025.11345072

79. RamojiRao, N., Vinodhini, G.A.F. (2025) Measuring the Efficiency of Random Forest, Naive Bayes, Multilayer Perceptron and Support Vector Machine in Email Spam Detection. *AIP Conference Proceedings*. https://doi.org/10.1063/5.0262529

80. Aguirre, A., Salazar, L. (2025) A Systematic Review of Artificial Intelligence Techniques for Phishing Detection. *Advances in Artificial Intelligence and Machine Learning*. https://doi.org/10.54364/aaiml.2025.53231

81. Kry, Y.E., Ravindran, D.J.S. (2025) A Hybrid Ensemble SMS Spam Detection System through Weighted Soft Voting. *2025 International Conference on Computational Intelligence and Knowledge Economy, ICCIKE 2025*. https://doi.org/10.1109/ICCIKE67021.2025.11318293

82. Kuanyshbay, D.N., Serek, A.G., Shoiynbek, A.A., (...), Meraliyev, M.A. (2025) Development of an AI-Based Communication Fraud Detection System. *Applied Mathematics and Information Sciences*. https://doi.org/10.18576/amis/190419

83. Yadav, A., Yadav, R., Kiruthika, M. (2025) SMS Spam Detection with NLP and Ensemble Learning for Enhanced Classification. *2025 International Conference on Computing Technologies and Data Communication, ICCTDC 2025*. https://doi.org/10.1109/ICCTDC64446.2025.11159039

84. Osa, E., Elaigwu, V.O. (2021) Modelling of a Deep Learning Based SMS Spam Detection Application. *NIPES - Journal of Science and Technology Research*. https://doi.org/10.37933/nipes/3.4.2021.17

85. Rashid, F., Doyle, B., Han, S.C., Seneviratne, S. (2024) Phishing URL detection generalisation using Unsupervised Domain Adaptation. *Computer Networks*. https://doi.org/10.1016/j.comnet.2024.110398

86. Awasthi, Anjaneya, Goel, Noopur (2022) Phishing website prediction using base and ensemble classifier techniques with cross-validation. *Cybersecurity*. https://doi.org/10.1186/s42400-022-00126-9

87. Hussain, M., Abbas, J., Hussain, J., (...), Wu, H. (2027) ExpertFusion: Calibrated multi-expert decision fusion for phishing URL detection under distribution shift and target-prior uncertainty. *Expert Systems with Applications*. https://doi.org/10.1016/j.eswa.2026.134089

88. Asif, T., Mehmood, F., Gillani, S.A.M., (...), Dengel, A. (2025) RPCP-PURI: A robust and precise computational predictor for Phishing Uniform Resource Identification. *Journal of Information Security and Applications*. https://doi.org/10.1016/j.jisa.2024.103953

89. Akanbi, Oluwatobi Ayodeji (2015) Literature Review. *A Machine-Learning Approach to Phishing Detection and Defense*. https://doi.org/10.1016/B978-0-12-802927-5.00002-2

90. Ma, J., Saul, L.K., Savage, S., Voelker, G.M. (2009) Beyond blacklists: Learning to detect malicious web sites from suspicious URLs. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. https://doi.org/10.1145/1557019.1557153

91. Zhang, Y., Hong, J.I., Cranor, L.F. (2007) Cantina: A content-based approach to detecting phishing web sites. *16th International World Wide Web Conference, WWW2007*. https://doi.org/10.1145/1242572.1242659

92. Xiang, G., Hong, J., Rosé, C.P., Cranor, L. (2011) CANTINA+: A feature-rich machine learning framework for detecting phishing web sites. *ACM Transactions on Information and System Security*. https://doi.org/10.1145/2019599.2019606

93. Garera, S., Provos, N., Chew, M., Rubin, A.D. (2007) A framework for detection and measurement of phishing attacks. *WORM'07 - Proceedings of the 2007 ACM Workshop on Recurring Malcode*. https://doi.org/10.1145/1314389.1314391

94. Ying, P., Ding, D. (2006) Anomaly based web phishing page detection. *Proceedings - Annual Computer Security Applications Conference, ACSAC*. https://doi.org/10.1109/ACSAC.2006.13

95. Kumar, P.V.P., Swetha, M., Kumari, P.M., (...), Harshitha, R. (2026) Explainable AI for Cyber Security: Interpretable Threat Analysis Using SHAP and LIME. *Proceedings of the 2026 13th International Conference on Computing for Sustainable Global Development, INDIACom 2026*. https://doi.org/10.23919/INDIACom70271.2026.11525454

96. Kaur, G., Kaur, G., Choudhury, A., (...), Srivastava, A. (2025) Explainable Artificial Intelligence in Cyber Defense-A Review. *2025 7th International Symposium on Advanced Electrical and Communication Technologies, ISAECT 2025*. https://doi.org/10.1109/ISAECT68904.2025.11318816

97. Rumapea, H., Manalu, D.R., Rumapea, Y.Y.P. (2025) Interpretable Deep Learning for Enhanced AI Trust and Clarity. *Journal of Artificial Intelligence and Technology*. https://doi.org/10.37965/jait.2025.0748

98. Pushkala, S.A. (2025) Explainable AI for Trustworthy Predictions: The Comparative Analysis of SHAP, LIME, and Counterfactuals. *ICRTEECT 2025 - 2nd International Conference on Recent Trends in Electrical, Electronics and Computing Technologies*. https://doi.org/10.1109/ICRTEECT67512.2025.11448658